



Exploring the shapes of graphs using technology

- 1 The linear equation $y = 3x - 7$ will form a _____ graph. (Tick **1** correct answer)

- ☐ curved
- ☐ decreasing
- ☐ straight line

- 2 Which of these coordinates will be on the linear graph $y = 3x - 7$? (Tick **3** correct answers)

- ☐ (5, 8)
- ☐ (8, 5)
- ☐ (-4, -1)
- ☐ (1, -4)
- ☐ (2, -1)

- 3 How many points can be plotted on the graph of the linear equation $y = 3x - 7$? (Tick **1** correct answer)

- ☐ All integers.
- ☐ Positive integer values only.
- ☐ Negative integer values only.
- ☐ Millions but, there is a limit.
- ☐ The number of points you can plot on this graph goes on infinitely.



- 4 If you were populating this table of values for the linear equation $y = 8 - 5x$, what is highest value of y you will get? (Tick **1** correct answer)

x	-3	-2	-1	0	1	2	3
y							

☐ 3

☐ 5

☐ 8

☐ 23

☐ 28

- 5 If you were populating this table of values for the linear equation $y = 8 - 5x$, what is lowest value of y you will get? (Tick **1** correct answer)

x	-3	-2	-1	0	1	2	3
y							

☐ 23

☐ -3

☐ -5

☐ -7

☐ -23

- 6 Which of these equation will produce a non-linear graph? (Tick **1** correct answer)

☐ $y = 3x + 8$
☐ $y = 3x - 8$
☐ $y = 8 - 3x$
☐ $3x + y = 8$
☐ $y = x^3 + 8$
